Answers Type : Equal
Text Areas : PlainText
Possible Answers :
1

Physics Section A

Section Id :	69112139
Section Number :	3
Section type :	Online
Mandatory or Optional :	Mandatory
Number of Questions :	20
Number of Questions to be attempted :	20
Section Marks :	80
Maximum Instruction Time :	0
Sub-Section Number :	1
Sub-Section Id :	69112139
Question Shuffling Allowed :	Yes
Is Section Default? :	No

Question Number : 26 Question Id : 691121476 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical

Match **List - I** with **List - II**.

	List - I		**List - II**
A.	Meter (L)	I.	$\sqrt{\frac{hc}{G}}$
B.	Second (S)	II.	$\sqrt{\frac{Gh}{c^5}}$
C.	Kilogram (M)	III.	$\sqrt{\frac{K^2L^2c^3}{Gh}}$
D.	Kelvin (K)	IV.	$\sqrt{\frac{Gh}{c^3}}$

where h (Planck's constant), G (gravitational constant) and c (speed of light in vacuum) as fundamental units.

Choose the **correct** answer from the options given below :

Options :

6911211616. A-II, B-IV, C-I, D-III

6911211617. A-IV, B-II, C-I, D-III

6911211618. A-IV, B-I, C-II, D-III

6911211619. A-III, B-I, C-II, D-IV

Question Number : 27 Question Id : 691121477 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical

In an experiment to determine the resistance of a given wire using Ohm's law, the voltmeter and ammeter readings are noted as 10 V and 5 A, respectively. The least counts of voltmeter and ammeter are 500 mV and 200 mA, respectively. The estimated error in the resistance measurement is _____ Ω

Options :

6911211620. 0.25

6911211621. 2

6911211622. 2.5

6911211623. 0.18

Question Number : 28 Question Id : 691121478 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical

A mass of 1 kg is kept on a inclined plane with 30° inclination with respect to horizontal plane and it is at rest initially. Then the whole assembly is moved up with constant velocity of 4 m/s. The work done by the frictional force in time 2 s is _____ J. (Take $g = 10\ m/s^2$)

Options :

6911211624. 20

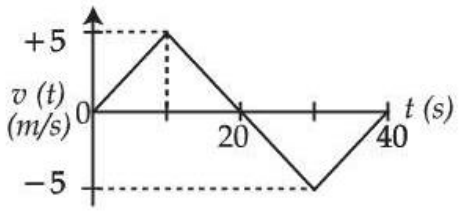
6911211625. 25

6911211626. 30

6911211627. 10

Question Number : 29 Question Id : 691121479 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical

The velocity (v) versus time (t) plot of a particle is shown in the figure, for a time interval of 40 s. The total distance travelled by the particle and the average velocity during this period are, respectively _____.



Options :

6911211628. 25 m and zero

6911211629. 50 m and zero

6911211630. 100 m and zero

6911211631. 100 m and 2.5 m/s

Question Number : 30 Question Id : 691121480 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical

A wheel initially at rest is subjected to a uniform angular acceleration about its axis. In the first 2 s it rotates through an angle θ_1 and in the next 2 s it rotates through an angle θ_2. The ratio $\frac{\theta_2}{\theta_1}$ is _____.

Options :

6911211632. 6

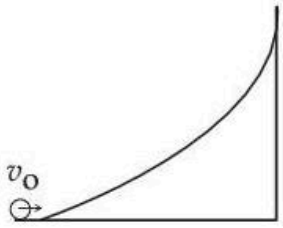
6911211633. 3

6911211634. 4

6911211635. $\frac{1}{3}$

Question Number : 31 Question Id : 691121481 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical

An object of uniform density rolls up the curved path with the initial velocity v_o as shown in the figure. If the maximum height attained by an object is $\frac{7v_o^2}{10\,g}$ (g = acceleration due to gravity), the object is a _____.



Options :

6911211636. solid cylinder

6911211637. ring

6911211638. disc

6911211639. solid sphere

Question Number : 32 Question Id : 691121482 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical

A body of mass m is taken from the surface of earth to a height equal to twice the radius of earth (R_e). The increase in potential energy will be _____.

(g is acceleration due to gravity at the surface of earth)

Options :

6911211640. $\frac{1}{2}mgR_e$

6911211641. $\frac{3}{4}mgR_e$

6911211642. $\frac{1}{4}mgR_e$

6911211643. $\frac{2}{3}mgR_e$

Question Number : 33 Question Id : 691121483 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical

Eight mercury drops, each of radius r, coalesce to form a bigger drop. The surface energy released in this process is _____. (S is the surface tension of mercury).

Options :

6911211644. $8\pi r^2 S$

6911211645. $16\pi r^2 S$

6911211646. $64\pi r^2 S$

6911211647. $4\pi r^2 S$

Question Number : 34 Question Id : 691121484 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical

An ideal gas at pressure P and temperature T is expanding such that $PT^3 = \text{constant}$. The coefficient of volume expansion of the gas is _____.

Options :

6911211648. $\frac{2}{T}$

6911211649. $\frac{1}{T}$

6911211650. $\frac{4}{T}$

6911211651. $\frac{3}{T}$

Question Number : 35 Question Id : 691121485 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical

Match **List - I** with **List - II**.

	List - I		**List - II**
A.	$\sin^2 \omega t$	I.	Periodic with time period $T=\frac{\pi}{\omega}$ but not simple harmonic motion (SHM)
B.	$\sin^3(2\omega t)$	II.	Periodic with time period $T=\frac{2\pi}{\omega}$ but Not SHM
C.	$\sin(\omega t)+\cos(\pi\omega t)$	III.	Periodic with time period $T=\frac{\pi}{\omega}$ and SHM
D.	$\cos \omega t+\cos 2\omega t$	IV.	Non-periodic

Choose the **correct** answer from the options given below :

Options :

6911211652. A-III, B-I, C-IV, D-II

6911211653. A-II, B-I, C-III, D-IV

6911211654. A-III, B-II, C-IV, D-I

6911211655. A-II, B-I, C-IV, D-III

Question Number : 36 Question Id : 691121486 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical

A metal rod of length L rotates about one end at origin with a uniform angular velocity ω. The magnetic field radially falls off as $B(r)=B_0 e^{-\lambda r}$; λ being a positive constant. The emf induced (neglecting the centripetal force on electrons in the rod) is :

Options :

6911211656. $B_0\omega\left[\frac{1}{\lambda^2}-e^{-\lambda L}\left(\frac{1}{\lambda^2}+\frac{L}{\lambda}\right)\right]$

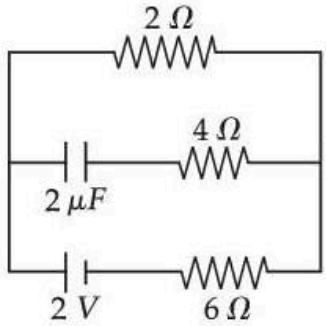
6911211657. $B_0\omega\left[\frac{1}{\lambda^2}+e^{-\lambda L}\left(\frac{1}{\lambda^2}+\frac{L}{\lambda}\right)\right]$

6911211658. $B_0\omega\left[\frac{4}{\lambda^2} - e^{-2\lambda L}\left(\frac{1}{\lambda^2} + \frac{2L}{\lambda}\right)\right]$

6911211659. $B_0\omega\left[\frac{3}{\lambda^2} - e^{-3\lambda L}\left(\frac{3}{\lambda^2} + \frac{L}{\lambda}\right)\right]$

Question Number : 37 Question Id : 691121487 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical

Under steady state condition the potential difference across the capacitor in the circuit is _____ V.



Options :

6911211660. 0.5

6911211661. 1.5

6911211662. 0

6911211663. 2

Question Number : 38 Question Id : 691121488 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical

A particle of charge q and mass m is projected from origin with an initial velocity $\vec{v} = \left(\frac{v_0}{\sqrt{2}}\hat{x} + \frac{v_0}{\sqrt{2}}\hat{y}\right)$. There exists a uniform magnetic field $\vec{B} = B_0\hat{z}$ and a space varying electric field $\vec{E} = E_0\, e^{-\lambda x}\hat{x}$ within the region $0 \leq x \leq L$. After travelling a distance such that x-coordinate has changed from $x = 0$ to $x = L$, the change in the kinetic energy is _____.

Options :

6911211664. $\frac{q\,E_0}{\lambda}\left[1 - e^{-\lambda L}\right]$

6911211665. $\left(\frac{v_0\, q\, B_0}{2\lambda}\right)\left[2 - e^{-2\lambda L}\right]$

6911211666. $\frac{q\,E_0}{\lambda}\left[1+e^{-\lambda L}\right]$

6911211667. $q\left(\frac{E_0+v_0\,B_0}{\lambda}\right)\left[1-e^{-\lambda L/2}\right]$

Question Number : 39 Question Id : 691121489 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical

Given below are two statements : one is labelled as **Assertion (A)** and the other is labelled as **Reason (R)**.

Assertion (A) : The electromagnetic wave exerts pressure on the surface on which they are allowed to fall.

Reason (R) : There is no mass associated with the electromagnetic waves.

In the light of the above statements, choose the **correct answer** from the options given below :

Options :

6911211668. Both **(A)** and **(R)** are true and **(R)** is the correct explanation of **(A)**

6911211669. Both **(A)** and **(R)** are true but **(R)** is **not** the correct explanation of **(A)**

6911211670. **(A)** is true but **(R)** is false

6911211671. **(A)** is false but **(R)** is true

Question Number : 40 Question Id : 691121490 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical

A thin convex lens and a thin concave lens are kept in contact and are co-axial. Which of the following statements is correct for this combination of two lenses ?

Options :

6911211672. behaves as concave lens if $|f_{convex}| > |f_{concave}|$

6911211673. behaves as concave lens if $|f_{convex}| < |f_{concave}|$

6911211674. behaves as convex lens if $|f_{convex}| > |f_{concave}|$

6911211675. Focal length of the lens system will change if the positions of two lenses are interchanged

Question Number : 41 Question Id : 691121491 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical

An object AB is placed 15 cm on the left of a convex lens P of focal length 10 cm. Another convex lens Q is now placed 15 cm right of lens P. If the focal length of lens Q is 15 cm, the final image is _____.

Options :

6911211676. virtual, formed at 7.5 cm right of lens Q, with a size bigger than that of AB

6911211677. real, formed at 7.5 cm right of lens Q, with a size same as that of AB

6911211678. formed at infinity.

6911211679. real, formed at 7 cm right of lens Q, with a size smaller than that of AB

Question Number : 42 Question Id : 691121492 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical

The maximum intensity in a Young's double slit experiment is I_o. Distance between the slits (d) is 5λ, where λ is the wavelength of light used. The intensity of the fringe, exactly opposite to one of the slits on the screen, placed at $D = 10d$ is _____.

Options :

6911211680. $\frac{I_o}{4}$

6911211681. $\frac{I_o}{2}$

6911211682. I_o

6911211683. $\frac{3I_o}{4}$

Question Number : 43 Question Id : 691121493 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical

An electron is travelling with a velocity v in free space and when it enters a medium, its velocity is reduced by 20%. The de Broglie wavelength of electron in the medium is $\alpha\lambda_0$, where λ_0 is its de Broglie wavelength in free space. The value of α is _____.

Options :

6911211684. 1.20

6911211685. 1.0

6911211686. 1.25

6911211687. 0.75

Question Number : 44 Question Id : 691121494 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical

Assuming the experimental mass of ${}^{12}_{6}C$ as 12 u, the mass defect of ${}^{12}_{6}C$ atom is _____MeV/c^2.

(Mass of proton = 1.00727 u. mass of neutron = 1.00866 u, 1 u = 931.5 MeV/c^2 and c is the speed of the light in vacuum).

Options :

6911211688. 127.5

6911211689. 89.03

6911211690. 272.0

6911211691. 92.0

Question Number : 45 Question Id : 691121495 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical

In a semiconductor p-n diode, the doping concentrations on p-side and n-side are 10^{15} atoms/cm^3 and 10^{18} atoms/cm^3, respectively. Which one of the following statements is true ?

Options :

6911211692. Widths of depletion region on either side of the interface are equal

6911211693. The depletion region width is more on p-side compared to that in n-side

6911211694. The depletion region width is more on n-side compared to that in p-side

6911211695. No depletion region forms because of unequal doping concentrations on p and n-sides

Physics Section B

Section Id :	69112140
Section Number :	4
Section type :	Online
Mandatory or Optional :	Mandatory
Number of Questions :	5
Number of Questions to be attempted :	5
Section Marks :	20
Maximum Instruction Time :	0
Sub-Section Number :	1
Sub-Section Id :	69112140
Question Shuffling Allowed :	Yes
Is Section Default? :	No

Question Number : 46 Question Id : 691121496 Question Type : SA Display Question Number : Yes Keyboard Layout : Inscript

A copper wire of length 3 m is stretched by 3 mm by applying an external force. The volume of the wire is 600×10^{-6} m^3. The elastic potential energy stored in the wire in stretched condition would be _____ J.

(Given Young modulus of copper $= 1.1 \times 10^{11}$ N/m^2)

Response Type : Numeric
Evaluation Required For SA : Yes
Show Word Count : Yes
Answers Type : Equal
Text Areas : PlainText
Possible Answers :
1

Question Number : 47 Question Id : 691121497 Question Type : SA Display Question Number : Yes Keyboard Layout : Inscript

The heat extracted out of x gram of water initially at 50 °C to cool it down to 0 °C is sufficient to evaporate $(1000 - x)$ gram of water also initially at 50 °C. The value of x (closest integer) is _____.

(Take latent heat of water 2256 kJ/kg. K, specific heat capacity of water 4200 J/kg. K)

Response Type : Numeric
Evaluation Required For SA : Yes
Show Word Count : Yes
Answers Type : Equal
Text Areas : PlainText
Possible Answers :
1

Question Number : 48 Question Id : 691121498 Question Type : SA Display Question Number : Yes Keyboard Layout : Inscript

A series LCR circuit with $R = 20\ \Omega$, $L = 1.6$ H and $C = 40\ \mu$F is connected to a variable frequency a.c. source. The inductive reactance at resonant frequency is _____ Ω.

Response Type : Numeric
Evaluation Required For SA : Yes
Show Word Count : Yes
Answers Type : Equal
Text Areas : PlainText
Possible Answers :
1

Question Number : 49 Question Id : 691121499 Question Type : SA Display Question Number : Yes Keyboard Layout : Inscript

When an external resistance of 5 Ω is connected across terminals of a cell, a current of 0.25 A flows through it. When the 5 Ω resistor is replaced by a 2 Ω resistor, a current of 0.5 A flows through it. The internal resistance of the cell is _____ Ω.

Response Type : Numeric
Evaluation Required For SA : Yes
Show Word Count : Yes
Answers Type : Equal
Text Areas : PlainText
Possible Answers :
1

Question Number : 50 Question Id : 691121500 Question Type : SA Display Question Number : Yes Keyboard Layout : Inscript